

SIMATS ENGINEERING

ASSESSMENT TOOL 2 – Level Exam 2

COURSE CODE: CSA0904

COURSE NAME: Programming in Java

COURSE OUTCOME COVERED

CO3: Exception Handling Fundamentals – Exception Types – Uncaught Exceptions – Try catch and finally paradigms – Built in Exceptions – User Defined Exceptions – Multithreaded Programming – Java Thread Model – Creating threads – Creating multiple threads – Thread priorities – Synchronization – Inter thread communication – Suspending, resuming and stopping threads.

(BL3,BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 4

Total Marks: 100

ANNEXURE A

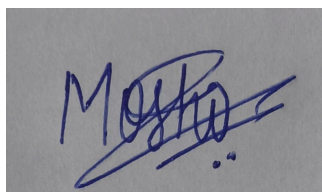
Level Exam-2

Code of Conduct:

I Moshika M - 192424064 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

A handwritten signature in blue ink, appearing to read 'Moshika', with a horizontal line drawn through it.

1. In an organization they decide to give bonuses to all the employees on New Year. A 5% bonus on salary is given to the grade A workers and 10% bonus on salary to the grade B workers. Write a program to enter the salary and grade of the employee. If the salary of the employee is less than \$10,000 then the employee gets an extra 2% bonus on salary. Calculate the bonus that has to be given to the employee and print the salary that the employee will get.

Sample Input & Output:

```
Enter the grade of the employee: B
Enter the employee salary: 50000
Salary=50000
Bonus=5000.0
Total to be paid:55000.0
```

ANSWER:

CODE

```
import java.util.Scanner;

class Employee {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        char grade = sc.next().charAt(0);
        double salary = sc.nextDouble();
        double bonus;
        if (grade == 'A' || grade == 'a')
            bonus = salary * 0.05;
        else if (grade == 'B' || grade == 'b')
            bonus = salary * 0.10;
        else
            bonus = 0;
        if (salary < 10000)
            bonus += salary * 0.02;
        System.out.println("Salary = " + salary);
        System.out.println("Bonus = " + bonus);
        System.out.println("Total to be paid : " + (salary + bonus));
    }
}
```

INPUT

```
B
50000
```

OUTPUT

```
Salary = 50000.0
Bonus = 5000.0
Total to be paid : 55000.0
```

2. Write a Program to Find the Nth Largest Number in an array.

Sample Input:

List : {14, 67, 48, 23, 5, 62}

N = 4

Sample Output:

4th Largest number: 23

ANSWER:

CODE

```
import java.util.*;
```

```
class NthLargest {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
        int size = sc.nextInt();  
        int[] arr = new int[size];  
        for (int i = 0; i < size; i++) {  
            arr[i] = sc.nextInt();  
        }  
        int n = sc.nextInt();  
        Arrays.sort(arr);  
        System.out.println(n + "th Largest number : " + arr[size - n]);  
    }  
}
```

INPUT

```
6  
14  
67  
48  
23  
5  
62  
4
```

OUTPUT

4th Largest number : 23

3. Write a program to read a character until a * is encountered. Also count the number of uppercase, lowercase, and numbers entered by the users.

Sample Input:

Enter * to exit...

Enter any character: W

Enter any character: d

Enter any character: A
Enter any character: G
Enter any character: g
Enter any character: H
Enter any character: *
Sample Output:
Total count of lower case:2
Total count of upper case:4
Total count of numbers =0

ANSWER:

CODE

```
import java.util.Scanner;
class CharacterCount {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int upper = 0, lower = 0, number = 0;
        char ch;
        while (true) {
            ch = sc.next().charAt(0);
            if (ch == '*')
                break;
            if (Character.isUpperCase(ch))
                upper++;
            else if (Character.isLowerCase(ch))
                lower++;
            else if (Character.isDigit(ch))
                number++;
        }
        System.out.println("Total count of lower case : " + lower);
        System.out.println("Total count of upper case : " + upper);
        System.out.println("Total count of numbers : " + number);
    }
}
```

INPUT

W
d
A
G
g
H
*

OUTPUT

Total count of lower case : 2

Total count of upper case : 4

Total count of numbers : 0

4. Write a program to print the first n perfect numbers. (Hint Perfect number means a positive integer that is equal to the sum of its proper divisors)

Sample Input:

N = 3

Sample Output:

First 3 perfect numbers are: 6 , 28 , 496

ANSWER:

CODE

```
import java.util.Scanner;
class PerfectNumbers {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int count = 0;
        int num = 2;
        System.out.print("First " + n + " perfect numbers are: ");
        while (count < n) {
            int sum = 0;
            for (int i = 1; i <= num / 2; i++) {
                if (num % i == 0)
                    sum += i;
            }
            if (sum == num) {
                System.out.print(num);
                count++;
                if (count < n)
                    System.out.print(" , ");
            }
            num++;
        }
    }
}
```

INPUT

3

OUTPUT

First 3 perfect numbers are: 6 , 28 , 496